Shiva Shankar S.

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Protein Biochemist & Molecular biologist

Protein biochemist with 15 years of experience in protein structure analysis, recombinant protein expression, proteomics, and protein crystallization. Skilled in structure-guided engineering, inverse folding, molecular cloning, protein purification, and biophysical characterization. Experienced in redesigning difficult proteins into functional biomedical variants, including a co-invented patent for a glycation-resistant insulin polypeptide. Currently based in Baltimore, MD, and seeking a postdoctoral position in protein design, protein engineering, and experimental characterization.

Core Competencies

- Protein engineering
- Recombinant Protein Expression
- Inverse Folding & Sequence Design
- MS-Based Proteomics (DDA/DIA/PRM)
- Molecular Cloning & Mutagenesis
- Cell-based functional assays
- Protein Crystallization
- Protein Purification & Characterization
- Teaching, Mentorship & Instruction

Professional Experience

CSIR-National Chemical Laboratory, Pune, India Jul 2017 - Dec 2025

Senior Research Fellow / Project Associate

Led independent research projects spanning computational protein design, mass spectrometry-based proteomics, and structural characterization. Developed and applied cutting-edge methods including inverse folding design, targeted proteomics assays, and bottom-up/top-down MS workflows.

Project 1: Meteorin & Meteorin-Like Protein Characterization

- First structural analysis of Meteorin family proteins using AlphaFold2/ColabFold and RoseTTAFold; identified functional receptor-binding regions and high-risk missense variants (*JBSD, 2023*)
- Applied ProteinMPNN-based inverse folding to redesign insoluble mouse Metrnl into a soluble, functionally active variant - manuscript in preparation (first & co-corresponding author)
- Developed targeted proteomics assay (PRM) for Metrnl quantification; mapped protein-protein interactions via XL-MS integrated with computational modelling

Project 2: Glycation-Resistant Insulin Engineering

- Engineered a novel glycation-resistant insulin proteoform via de novo solubility-tag design; co-invented Patent WO 2024/194897 A1 (issued Sep 2024)
- Designed novel in vitro workflow for studying insulin glycation mechanisms; characterized modification profiles by bottom-up and top-down MS
- Optimized recombinant expression and inclusion body refolding systems, achieving significant yield improvements

Project 3: SARS-CoV-2 Main Protease Inhibitor Discovery

- Contributed computational structural analysis and molecular docking to identify multi-target-directed ligands against SARS-CoV-2 Mpro (*JBSD, 2021*)

CSIR-National Chemical Laboratory, Pune, India Apr 2013 - Jul 2017

Project Assistant II

Performed structural and functional characterization of Santalene synthase from Santalum album (Indian sandalwood), contributing to the discovery of novel sesquiterpene synthase activities.

- Executed cloning, mutagenesis, multi-step purification, crystallization, and functional characterization including kinetics and product identification (*Sci. Rep., 2015*)
- Gained direct experience in structural and mechanistic analysis of biosynthetic enzymes, including active site mapping and structure-function relationships

IISER-Pune, India Jul 2010 - Apr 2013

Junior Research Fellow

Investigated HIV-1 fusion inhibitors and self-assembling antimicrobial lipopeptides, contributing to fundamental research in peptide-based therapeutics.

- Synthesized and characterized gamma- and hybrid gamma-peptide HIV-1 fusion inhibitors and antimicrobial lipopeptides (first-author, *J. Med. Chem., 2013*)

National Institute of Immunology (NII), New Delhi, India Jul 2009 - Apr 2010

Project Trainee

- Cloned, expressed, and purified HisA, HisB, and HisC2 proteins from *M. tuberculosis* histidine biosynthesis pathway for crystallographic studies

Education

Ph.D. in Biological Sciences, CSIR-National Chemical Laboratory, Pune, India Jul 2017 - Dec 2025
Thesis: Structural and functional characterization of Meteorin and Meteorin-like proteins
Advisors: Dr. Mahesh J. Kulkarni & Dr. Sureshkumar Ramasamy
M.Sc. in Biotechnology, Bharathiar University, India - 2009
B.Sc. in Biotechnology, Bharathiar University, India - 2007

Awards & Recognition

3rd Prize-Basic Science, Chellaram Foundation Diabetes Research Award | 7th IDS (Diabetes summit), Pune **2023**
Travel Award | 11th Annual Meeting of Proteomics Society of India **2019**
CSIR Senior Research Fellowship - Competitive national fellowship for doctoral research **2017**
GATE Qualified - Biotechnology, All India Rank 862 **2016**
Summer Research Fellowship - Indian Academy of Sciences **2008**

Teaching & Mentorship

CSIR Integrated Skill Initiative 2022
Instructor - Targeted Proteomics Workshop
Designed and delivered curriculum on PRM assay development and data analysis using Skyline; trained researchers from national institutes.

CSIR Integrated Skill Initiative 2021
Instructor - Basic to Advanced Proteomics Course
Led hands-on sessions from sample preparation to data interpretation using MaxQuant and FragPipe.

CSIR-National Chemical Laboratory 2017 - 2025
Student Mentorship
Supervised 8 project students and trained 4 Ph.D. students in molecular biology, protein purification, protein design, and proteomics workflows.
Scientific Peer Review: Analytical Biochemistry; Protein Expression and Purification; ACS Omega

Scientific Publications

Patent

- Shankar, S.S.*, Kulkarni, M.J., Banarjee, R., Jathar, S., Rajesh, S. A glycation-resistant insulin polypeptide and a method of preparing the same thereof. WO 2024/194897 A1 · Issued Sep 26, 2024

Peer-Reviewed Articles (* First and Co-corresponding author)

2025 Rajesh, S., Jathar, S., Banarjee, R., Sharma, M., Palkar, S., Shankar, S.S.*, & Kulkarni, M.J. A simple freeze-thaw based method for efficient purification of recombinant human proinsulin from inclusion bodies. *Protein Expression and Purification,* 227: 106645.

2023 Shankar, S.S.*, Banarjee, R., Jathar, S.M., Rajesh, S., Ramasamy, S., & Kulkarni, M.J. De novo Structure Prediction of Meteorin and Meteorin-Like Protein for Identification of Domains, Functional Receptor Binding Regions, and High-Risk Missense Variants. *Journal of Biomolecular Structure and Dynamics,* 42(9): 4522-4536.

2023 Pandya, V.K., Shankar, S.S., Sonwane, B.P., Rajesh, S., Rathore, R., et al. Mechanistic insights on anserine hydrolyzing activities of human carnosinase. *Biochimica et Biophysica Acta (BBA) - General Subjects,* 1867(3): 130290.

2021 Joshi, R.S., Jagdale, S.S., Bansode, S.B., Shankar, S.S., et al. Discovery of potential multi-target-directed ligands by targeting host-specific SARS-CoV-2 structurally conserved main protease. *Journal of Biomolecular Structure and Dynamics,* 39(9): 3099-3114.

2015 Srivastava, P.L., Daramwar, P.P., Krithika, R., Pandreka, A., Shankar, S.S., & Thulasiram, H.V. Functional Characterization of Novel Sesquiterpene Synthases from Indian Sandalwood, Santalum album. *Scientific Reports,* 5: 10095.

2013 Shankar, S.S.*, Benke, S.N., Nagendra, N., Srivastava, P.L., Thulasiram, H.V., & Gopi, H.N. Self-Assembly to Function: Design, Synthesis, and Broad-Spectrum Antimicrobial Properties of Short Hybrid E-Vinylogous Lipopeptides. *Journal of Medicinal Chemistry,* 56(21): 8468-8474.

Manuscripts in Preparation

- Shankar, S.S.*, Jathar, S., Sharma, M., Dole, A., Rajesh, S., Banarjee, R., Kulkarni, M.J. Inverse Folding Design Enables Soluble Expression and Functional Activity of Mouse Meteorin-Like Protein.
- Jathar, S., Dole, A., Rajesh, S., Shankar, S.S.*, Kulkarni, M.J. Characterization of Glycation-Modified Insulin Glargine by Top-Down Mass Spectrometry.

• Shankar, S.S.*, Jathar, S.M., & Kulkarni, M.J. Structure-Guided Discovery of RAGE V-Domain Antagonists Targeting AGE- and Ligand-Mediated Inflammaging.

Conference & Poster Presentations

- 54th American Aging association-AGE meeting, Utah, USA - Jun 2026
- 73rd ASMS Conference on Mass Spectrometry, Baltimore, USA - Jun 2025
- 21st Annual US HUPO Conference, Philadelphia, USA - Feb 2025
- Advanced Proteomics Technologies Workshop, IIT Bombay - Feb 2024
- Monsoon Advanced Proteomics School (MAPS), IIT Bombay - Jun 2023
- 7th International Diabetes Summit, Pune - Mar 2023
- 11th Annual Meeting of Proteomics Society of India - 2019
- 24th IUCr Congress of Crystallography, India - Aug 2017

Technical Proficiencies

Proteomics & Mass Spectrometry: Bottom-up and top-down proteomics, DDA, DIA/SWATH, PRM workflows, targeted assay development, XL-MS for interaction mapping, PTM enrichment & analysis (phosphorylation, glycation), IP-MS, MALDI, MaxQuant, Skyline, FragPipe, R, Python

Protein Expression, Purification & Characterization: Recombinant expression (*E. coli*, insect cells, mammalian cells), molecular cloning & site-directed mutagenesis, affinity/ion-exchange/gel-filtration chromatography, HPLC, FPLC (AKTA), Inclusion body refolding, SEC-MALS, DLS, circular dichroism, ITC, AFM

Molecular & Cell Biology: Mammalian cell culture, CRISPR-based knockout generation, immunofluorescence, cell-based assays, PCR / qPCR / RT-PCR, Western blotting, ELISA

Structural Biology & Computational Design: Protein crystallization (hanging-drop, sitting-drop, vapor diffusion), robotic crystallization (Hamilton, Mosquito), X-ray diffraction, AlphaFold2/ColabFold, RoseTTAFold, inverse folding & sequence design (ProteinMPNN, RFdiffusion, Bindcraft), homology modelling (Modeller, Schrodinger Prime), molecular dynamics (GROMACS, Desmond), protein-ligand docking, PyMOL, ChimeraX

References

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